

Erica: AI Tutor

You: What is attention in transformers and can you provide a python example of how it is used?

Erica:

Based on our course material:

Attention Mechanism in Transformers

The **attention mechanism** is a key component in transformer models, enabling them to focus on relevant parts of the input sequence when generating an output. The attention mechanism allows each token (or element) in the input sequence to consider every other token's contribution.

Prerequisites

1. Self-Attention Mechanism

- Definition: A form of attention where the queries, keys, and values come from the same input sequence.
- Mathematical Formula: $\text{Attention}(Q, K, V) = \text{softmax}((QK^T)/(\sqrt{d_k}))V$ Where:
- Q : Query matrix of shape (N, d_k)
- K : Key matrix of shape (N, d_k)

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- Mathematical Formula: $\text{Attention}(Q, K, V) = \text{softmax}((QK^T)/(\sqrt{d_k}))V$ Where:
- Q : Query matrix of shape (N, d_k)
- K : Key matrix of shape (N, d_k)
- V : Value matrix of shape (N, d_v)
- d_k : Dimension of the key vector
- softmax is applied row-wise

2. Multi-Head Attention

- Definition: An attention mechanism that runs multiple self-attention operations in parallel.
- This helps capture diverse relationships and patterns across the input sequence.

3. Attention Score Scaling

- Definition: The practice of dividing the dot-product attention score by the square root of the key

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3. Attention Score Scaling

- Definition: The practice of dividing the dot-product attention score by the square root of the key dimension to prevent extremely large values that destabilize the softmax function.

4. Dot-Product Attention

- Definition: A mechanism that computes attention weights by taking the dot product between a query vector and key vectors, scaling the result, applying softmax, and weighting the value vectors accordingly.
- Mathematical Formula: $\text{Attention}(Q, K, V) = \text{softmax}((QK^T)/(\text{sqrt}(d_k)))V$

Example of Attention Mechanism in Python (Simplified)

```
import torch
from torch import nn

class SimpleSelfAttention(nn.Module):
    def __init__(self, d_model):
        super(SimpleSelfAttention, self).__init__()
        self.d_model = d_model
```

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```
import torch
from torch import nn

class SimpleSelfAttention(nn.Module):
    def __init__(self, d_model):
        super(SimpleSelfAttention, self).__init__()
        self.d_model = d_model

    def forward(self, x):
        # x: [batch_size, seq_length, d_model]

        # Compute Query (Q), Key (K), and Value (V)
        Q = x # Assuming the query is the same as input
        K = x # Assuming the key is the same as input
        V = x # Assuming the value is the same as input

        # Compute Attention Scores
        attention_scores = torch.matmul(Q, K.transpose(-2, -1)) / torch.sqrt(torch.tensor(self.d_model).float())
```

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```
# Apply Softmax to normalize scores
attention_weights = nn.functional.softmax(attention_scores, dim=-1)

# Compute the weighted sum of values using the attention weights
output = torch.matmul(attention_weights, V)

return output

# Example usage:
batch_size = 2
seq_length = 5
d_model = 32

# Random input tensor [batch_size, seq_length, d_model]
x = torch.randn(batch_size, seq_length, d_model)

# Initialize the self-attention layer
self_attention_layer = SimpleSelfAttention(d_model)

# Apply the attention mechanism
```

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```
# Initialize the self-attention layer
self_attention_layer = SimpleSelfAttention(d_model)

# Apply the attention mechanism
output = self_attention_layer(x)
print("Output shape:", output.shape) # Expected: [2, 5, 32]
```

In this example:

- We define a simple **self-attention** layer where Q, K, and V are all derived from the same input sequence.
- The attention scores are computed using the dot product between queries and keys, normalized by the square root of the key dimension ($\sqrt{d_k}$).
- The softmax function is applied to these scores to get the attention weights.
- Finally, we compute a weighted sum of values using these attention weights.

This example provides a basic understanding of how self-attention works in transformers. In practice, multi-head attention (MHA) is often used for better performance and capturing different types of relationships within the input sequence.

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FileEditSelectionViewGoRunTerminalHelp

AI_Project

EXPLORER

AI_PROJECT

> __pycache__

> .env

> artifacts

≡ chunks.pkl

≡ graph.pkl

> data

> processed

> raw

> vector_db

≡ ingested_urls.txt

{ } kg_course.json

> erica_graph_rag_notebooks

> raw_data

01_scrape_data.ipynb

02_build_graph - Copy.i...

02_build_graph.ipynb

≡ dummy_knowledge_gra...

≡ knowledge_graph - Cop...

≡ knowledge_graph.gpickle

≡ knowledge_graph.pkl

≡ knowledge_graph1.pkl

≡ scraped_content.txt

> nano-graphrag

> pages

> static

> templates

⚙ .env

🔗 app.py

🔗 app1.py

🔗 build_vector_db.py

> OUTLINE

> TIMELINE

app1.py

01_scrape_data.ipynb

erica_graph_rag_notebooks

kg_q2.png

full_code_new.py

full_code_new1.py

latest_graphrag_front.html

old_app_f

app1.py > format_llm_output

111 def build_prompt(query, nodes):

143 {context_text}

144

145 QUESTION:

146 {query} """

147

148 print("Prompt Sent to Model:")

PROBLEMS 7

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

JUPYTER

=====

127.0.0.1 - - [05/Dec/2025 20:41:35] "POST /ask HTTP/1.1" 200 -

Inside answer def

retrieve relevant thingy

Retrieved KG Nodes:

['Self-Attention Mechanism', 'Multi-Head Attention', 'Attention Score Scaling', 'Dot-Product Attention', 'Scene Understanding']

=====

Saved KG subgraph visualization at static/query_graph.png

built prompt!

Prompt Sent to Model:

You are Erica, the AI tutor for Prof. Pantelis Monogioudis's "Introduction to AI" course.

RULES:

1. First use the course material below if relevant

2. If not enough in course material, add basic general knowledge

3. Always show any mathematical formulas or notation that exist for the topic

4. Keep explanations simple and clear

5. When explaining a concept, FIRST explain its prerequisites and show their mathematical formulas

6. Build up understanding step-by-step from foundational concepts to the current concept

WHEN ANSWERING:

- Start with: "Based on our course material"

- If adding info: "In general AI"

- Always show formulas: If topic has math, display it with \$\$like this\$\$

- For prerequisites: Explain each prerequisite concept with its own mathematical notation before connecting them to the main concept

COURSE MATERIAL:

Concept: Self-Attention Mechanism

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powershell

Python

```
app1.py > format_llm_output
111 def build_prompt(query, nodes):
112     {context_text}
144
145     QUESTION:
146     {query} """
147
148     print("Prompt Sent to Model:")
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

COURSE MATERIAL:

Concept: Self-Attention Mechanism

Aliases: self-attention, intra-attention

Definition: A form of attention where the queries, keys, and values come from the same input sequence, allowing each token to dynamically weigh the importance of every other token.

Concept: Multi-Head Attention

Aliases: MHA, parallel attention heads

Definition: An attention mechanism that runs multiple self-attention operations in parallel, enabling the model to capture diverse relationships and patterns across the input sequence.

Concept: Attention Score Scaling

Aliases: scaled attention, temperature scaling

Definition: The practice of dividing the dot-product attention score by the square root of the key dimension to prevent extremely large values that destabilize the softmax function.

Concept: Dot-Product Attention

Aliases: scaled dot-product attention, self-attention core

Definition: A mechanism that computes attention weights by taking the dot product between a query vector and key vectors, scaling the result, applying softmax, and weighting the value vectors accordingly.

Concept: Scene Understanding

Aliases: scene analysis, image understanding

Definition: The process of analyzing an image or scene to understand the objects present, their relationships, and how they interact. It involves not only recognizing individual objects but also understanding the context and layout.

QUESTION:

What is attention in transformers and can you provide a python example of how it is used?